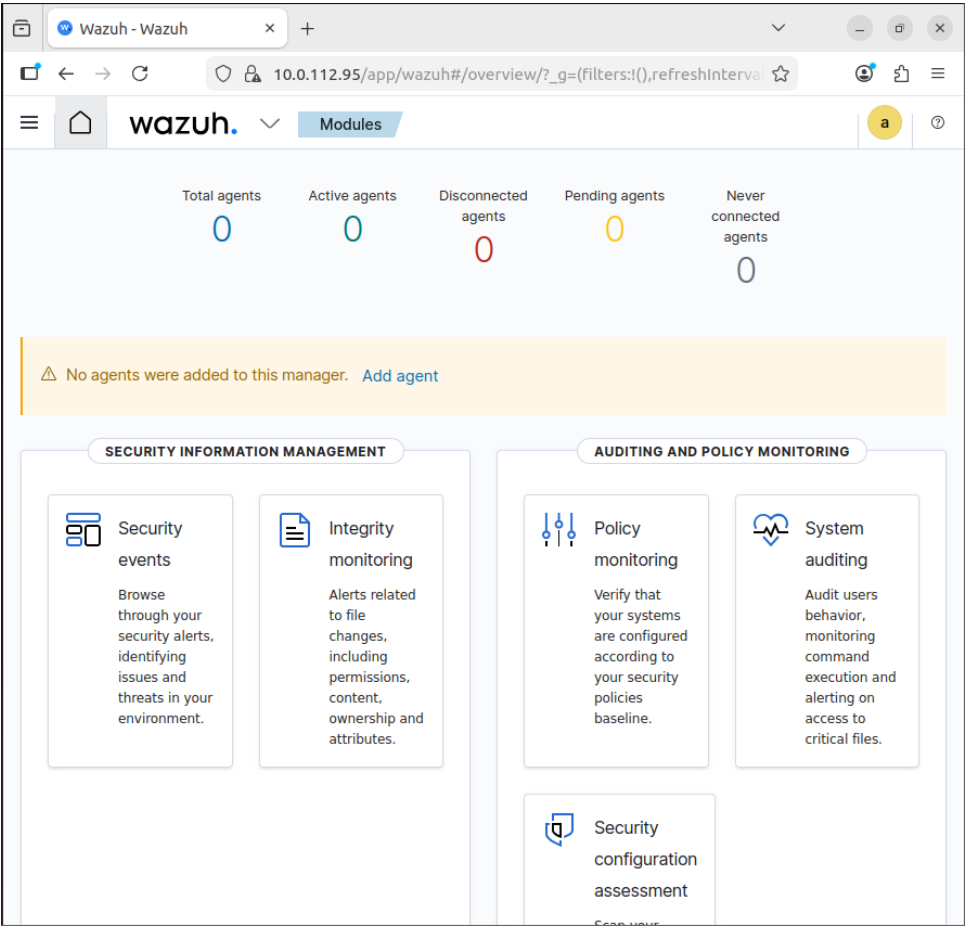


WAZUH INSTALLED

ACCESS WAZUH DASHBOARD



PART 2 — INSTALL WAZUH AGENT ON PLC VM

STEP 7 — INSTALL AGENT

WAZUH AGENT IS RUNNING ON THE PLC VM

```
test@OpenPLC-VM:~$ sudo systemctl status wazuh-agent
● wazuh-agent.service - Wazuh agent
   Loaded: loaded (/lib/systemd/system/wazuh-agent.service; enabled; vendor preset: enabled)
   Active: active (running) since Fri 2026-04-24 21:44:57 IST; 12s ago
     Process: 5993 ExecStart=/usr/bin/env /var/ossec/bin/wazuh-control start (code=exited, status=0/SUCCESS)
    Tasks: 30 (limit: 3465)
   Memory: 59.3M
      CPU: 6.152s
   CGroup: /system.slice/wazuh-agent.service
           └─6015 /var/ossec/bin/wazuh-execd
             └─6023 /var/ossec/bin/wazuh-agentd
               └─6036 /var/ossec/bin/wazuh-syscheckd
                 └─6049 /var/ossec/bin/wazuh-logcollector
                   └─6066 /var/ossec/bin/wazuh-modulesd

Apr 24 21:44:50 OpenPLC-VM systemd[1]: Starting Wazuh agent...
Apr 24 21:44:50 OpenPLC-VM env[5993]: Starting Wazuh v4.14.5...
Apr 24 21:44:50 OpenPLC-VM env[5993]: Started wazuh-execd...
Apr 24 21:44:51 OpenPLC-VM env[5993]: Started wazuh-agentd...
Apr 24 21:44:52 OpenPLC-VM env[5993]: Started wazuh-syscheckd...
Apr 24 21:44:54 OpenPLC-VM env[5993]: Started wazuh-logcollector...
Apr 24 21:44:55 OpenPLC-VM env[5993]: Started wazuh-modulesd...
Apr 24 21:44:57 OpenPLC-VM env[5993]: Completed.
Apr 24 21:44:57 OpenPLC-VM systemd[1]: Started Wazuh agent.
lines 1-23/23 (END)
```

```

test@OpenPLC-VM:~$ sudo systemctl restart wazuh-agent
[sudo] password for test:
test@OpenPLC-VM:~$ sudo systemctl status wazuh-agent
● wazuh-agent.service - Wazuh agent
   Loaded: loaded (/lib/systemd/system/wazuh-agent.service; enabled; vendor preset: enabled)
   Active: active (running) since Fri 2026-04-24 21:48:51 IST; 6s ago
     Process: 4802 ExecStart=/usr/bin/env /var/ossec/bin/wazuh-control start (code=exited, status=0/SUCCESS)
    Tasks: 30 (limit: 3465)
   Memory: 20.1M
     CPU: 5.899s
   CGroup: /system.slice/wazuh-agent.service
           └─4824 /var/ossec/bin/wazuh-execd
             └─4835 /var/ossec/bin/wazuh-agentd
               └─4849 /var/ossec/bin/wazuh-syscheckd
                 └─4862 /var/ossec/bin/wazuh-logcollector
                   └─4879 /var/ossec/bin/wazuh-modulesd

Apr 24 21:48:44 OpenPLC-VM systemd[1]: Starting Wazuh agent...
Apr 24 21:48:44 OpenPLC-VM env[4802]: Starting Wazuh v4.14.5...
Apr 24 21:48:46 OpenPLC-VM env[4802]: Started wazuh-execd...
Apr 24 21:48:47 OpenPLC-VM env[4802]: Started wazuh-agentd...
Apr 24 21:48:48 OpenPLC-VM env[4802]: Started wazuh-syscheckd...
Apr 24 21:48:49 OpenPLC-VM env[4802]: Started wazuh-logcollector...
Apr 24 21:48:49 OpenPLC-VM env[4802]: Started wazuh-modulesd...
Apr 24 21:48:51 OpenPLC-VM env[4802]: Completed.
Apr 24 21:48:51 OpenPLC-VM systemd[1]: Started Wazuh agent.
lines 1-23/23 (END)

```

```

test@OpenPLC-VM:~$ wget https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.7.5-1_amd64.deb
--2026-04-24 22:05:44-- https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.7.5-1_amd64.deb
Resolving packages.wazuh.com (packages.wazuh.com)... 99.86.20.122, 99.86.20.52, 99.86.20.28, ...
Connecting to packages.wazuh.com (packages.wazuh.com)|99.86.20.122|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 9378818 (8.9M) [application/vnd.debian.binary-package]
Saving to: 'wazuh-agent_4.7.5-1_amd64.deb'

wazuh-agent_4.7.5-1_amd 100%[=====>] 8.94M 6.46MB/s in 1.4s

2026-04-24 22:05:46 (6.46 MB/s) - 'wazuh-agent_4.7.5-1_amd64.deb' saved [9378818/9378818]

```

```
test@OpenPLC-VM:~$ sudo systemctl status wazuh-agent
● wazuh-agent.service - Wazuh agent
   Loaded: loaded (/lib/systemd/system/wazuh-agent.service; enabled; vendor preset: enabled)
   Active: active (running) since Fri 2026-04-24 22:28:32 IST; 6s ago
 Process: 4531 ExecStart=/usr/bin/env /var/ossec/bin/wazuh-control start (code=exited, status=0/SUCCESS)
    Tasks: 30 (limit: 3465)
   Memory: 134.1M
      CPU: 8.909s
   CGroup: /system.slice/wazuh-agent.service
           └─4553 /var/ossec/bin/wazuh-execd
             └─4564 /var/ossec/bin/wazuh-agentd
               └─4577 /var/ossec/bin/wazuh-syscheckd
                 └─4590 /var/ossec/bin/wazuh-logcollector
                   └─4598 /var/ossec/bin/wazuh-modulesd

Apr 24 22:28:25 OpenPLC-VM systemd[1]: Starting Wazuh agent...
Apr 24 22:28:25 OpenPLC-VM env[4531]: Starting Wazuh v4.7.5...
Apr 24 22:28:27 OpenPLC-VM env[4531]: Started wazuh-execd...
Apr 24 22:28:28 OpenPLC-VM env[4531]: Started wazuh-agentd...
Apr 24 22:28:29 OpenPLC-VM env[4531]: Started wazuh-syscheckd...
Apr 24 22:28:29 OpenPLC-VM env[4531]: Started wazuh-logcollector...
Apr 24 22:28:30 OpenPLC-VM env[4531]: Started wazuh-modulesd...
Apr 24 22:28:32 OpenPLC-VM env[4531]: Completed.
Apr 24 22:28:32 OpenPLC-VM systemd[1]: Started Wazuh agent.
test@OpenPLC-VM:~$ ping 10.0.112.95
PING 10.0.112.95 (10.0.112.95) 56(84) bytes of data:
64 bytes from 10.0.112.95: icmp_seq=1 ttl=64 time=1.38 ms
64 bytes from 10.0.112.95: icmp_seq=2 ttl=64 time=0.898 ms
64 bytes from 10.0.112.95: icmp_seq=3 ttl=64 time=1.04 ms
64 bytes from 10.0.112.95: icmp_seq=4 ttl=64 time=0.883 ms
64 bytes from 10.0.112.95: icmp_seq=5 ttl=64 time=0.841 ms
```

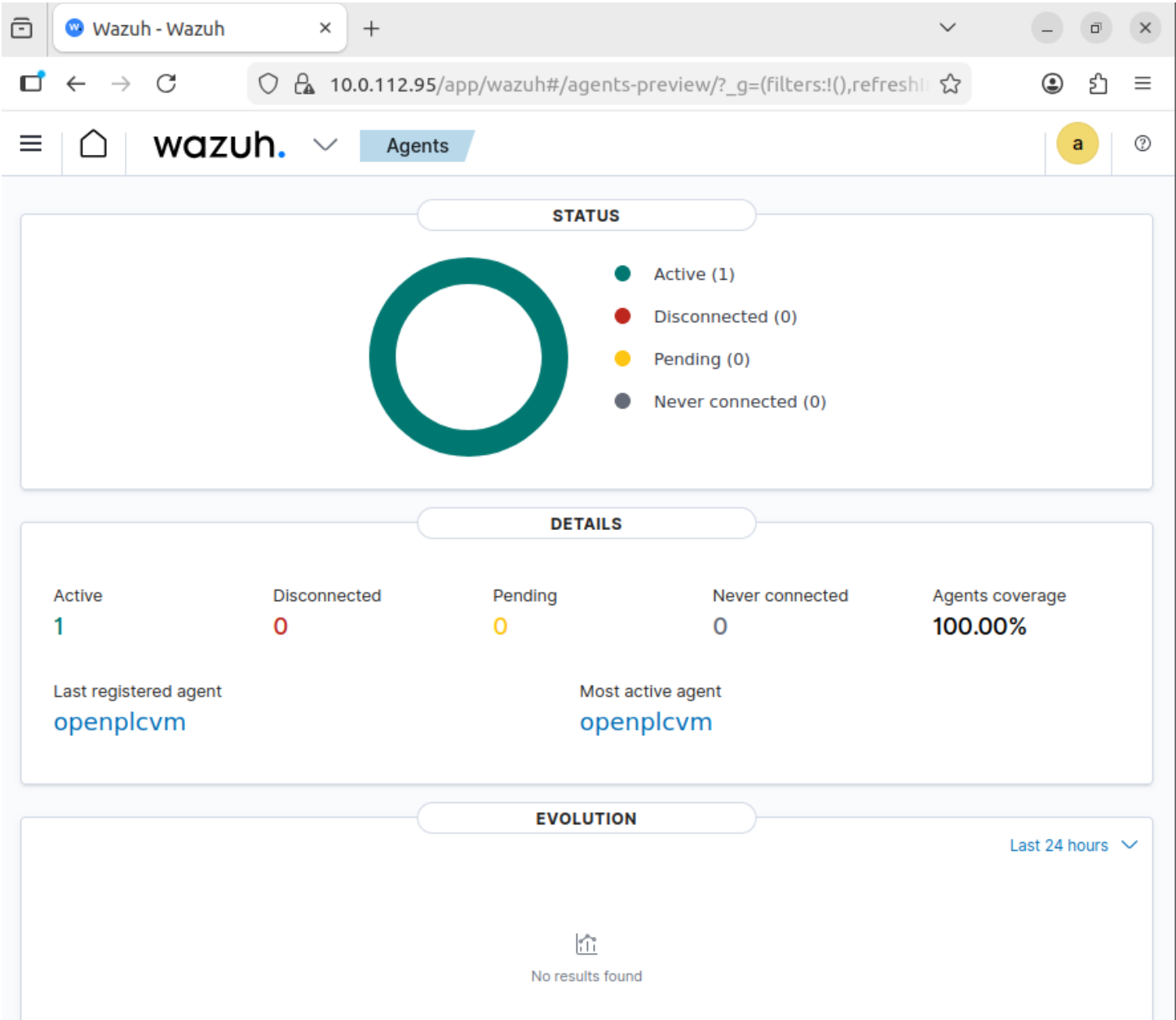
```
test@OpenPLC-VM:~$ sudo /var/ossec/bin/agent_control -l

Wazuh agent_control. List of available agents:
  ID: 000, Name: OpenPLC-VM (server), IP: 127.0.0.1, Active/Local
  ID: 001, Name: openplcvm, IP: any, Never connected
```

```
test@OpenPLC-VM:~$ sudo /var/ossec/bin/agent_control -l

Wazuh agent_control. List of available agents:
  ID: 000, Name: OpenPLC-VM (server), IP: 127.0.0.1, Active/Local
  ID: 001, Name: openplcvm, IP: any, Active

List of agentless devices:
```



CONNECTING WITH SURICATA

```
test@OpenPLC-VM:~$ sudo systemctl status suricata
● suricata.service - Suricata IDS/IDP daemon
   Loaded: loaded (/lib/systemd/system/suricata.service; enabled; vendor
   Active: active (running) since Fri 2026-04-24 22:44:06 IST; 4min 2s
     Docs: man:suricata(8)
           man:suricatasc(8)
           https://suricata-ids.org/docs/
   Process: 1035 ExecStart=/usr/bin/suricata -D --af-packet -c /etc/surica
 Main PID: 1229 (Suricata-Main)
    Tasks: 9 (limit: 3465)
   Memory: 66.3M
      CPU: 38.048s
   CGroup: /system.slice/suricata.service
           └─1229 /usr/bin/suricata -D --af-packet -c /etc/suricata/sur

Apr 24 22:44:04 OpenPLC-VM systemd[1]: Starting Suricata IDS/IDP daemon...
Apr 24 22:44:05 OpenPLC-VM suricata[1035]: 24/4/2026 -- 22:44:05 - <Notic
Apr 24 22:44:06 OpenPLC-VM systemd[1]: Started Suricata IDS/IDP daemon.
```

STEP 2: Check Suricata logs (VERY IMPORTANT)

```
test@OpenPLC-VM:~$ ls /var/log/suricata/
eve.json          fast.log          stats.log          suricata.log
eve.json.1.gz     fast.log.1.gz     stats.log.1.gz     suricata.log.1.gz
eve.json.2.gz     fast.log.2.gz     stats.log.2.gz     suricata.log.2.gz
test@OpenPLC-VM:~$
```

Now check live logs:

```
test@OpenPLC-VM:~$ sudo tail -f /var/log/suricata/eve.json
[sudo] password for test:
{"timestamp":"2026-04-24T22:50:50.344669+0530","flow_id":1924341388974064,
"in_iface":"ens33","event_type":"flow","src_ip":"10.0.112.30","src_port":6
0801,"dest_ip":"142.250.206.10","dest_port":443,"proto":"UDP","app_proto":
"failed","flow":{"pkts_toserver":14,"pkts_toclient":16,"bytes_toserver":42
72,"bytes_toclient":5169,"start":"2026-04-24T22:44:18.235504+0530","end":
"2026-04-24T22:44:41.062564+0530","age":23,"state":"established","reason":
"timeout","alerted":false}}
{"timestamp":"2026-04-24T22:50:51.011166+0530","flow_id":2065431078081888,
"in_iface":"ens33","event_type":"flow","src_ip":"10.0.112.50","src_port":6
4471,"dest_ip":"239.255.255.250","dest_port":1900,"proto":"UDP","app_proto
":"failed","flow":{"pkts_toserver":4,"pkts_toclient":0,"bytes_toserver":84
8,"bytes_toclient":0,"start":"2026-04-24T22:47:43.562528+0530","end":"2026
-04-24T22:47:46.589823+0530","age":3,"state":"new","reason":"timeout","ale
rted":false}}
{"timestamp":"2026-04-24T22:50:52.413312+0530","flow_id":1925239043449822,
"in_iface":"ens33","event_type":"flow","src_ip":"10.0.112.30","src_port":6
2505,"dest_ip":"172.217.24.206","dest_port":443,"proto":"TCP","app_proto":
"tls","flow":{"pkts_toserver":12,"pkts_toclient":7,"bytes_toserver":2718,"
bytes_toclient":5941,"start":"2026-04-24T22:45:54.254942+0530","end":"2026
-04-24T22:47:47.826560+0530","age":113,"state":"closed","reason":"timeout
","alerted":false},"tcp":{"tcp_flags":"1b","tcp_flags_ts":"1b","tcp_flags_t
c":"1b","syn":true,"fin":true,"psh":true,"ack":true,"state":"closed"}}
```

STEP 3: Configure Wazuh Agent to read Suricata logs

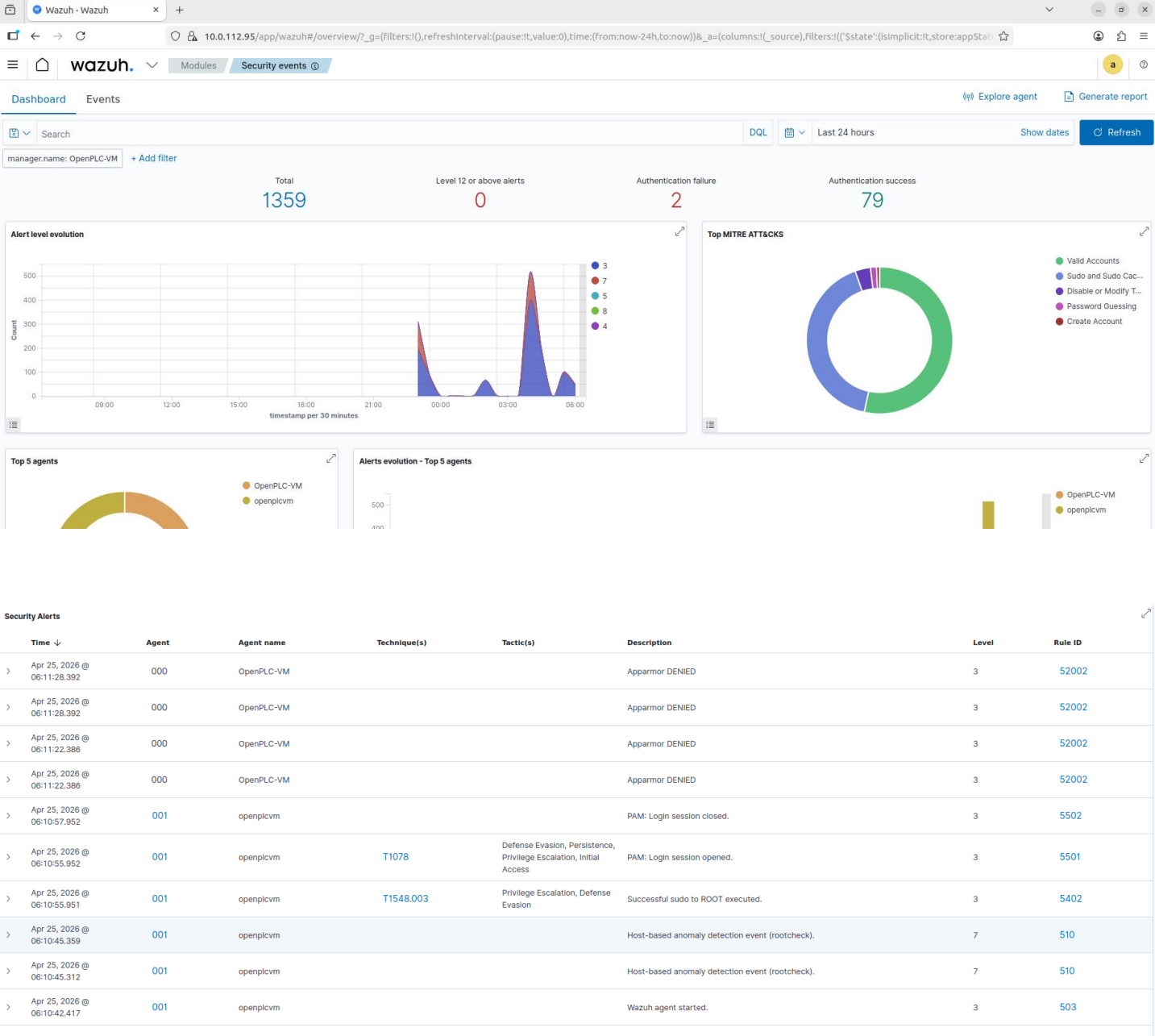
```
<localfile>
<log_format>json</log_format>
  <location>/var/log/suricata/eve.json</location>
</localfile>
</ossec_config>
```

STEP 4: Restart Wazuh Agent

STEP 5: Verify logs are sent

```
test@OpenPLC-VM:~$ sudo tail -f /var/ossec/logs/ossec.log
2026/04/24 22:56:30 wazuh-logcollector: INFO: Monitoring full output of co
mmand(360): netstat -tulpn | sed 's/\[[[:alnum:]]\+\]\ \+[\[[[:digit:]]\+\] \+
[\[[[:digit:]]\+\] \+\.*)\:\[\[[[:digit:]]*\]\ \+\([0-9\.\:*\]\+\)\.\+\ \[\[[[:dig
it:]]*\]\ \+[\[[[:alnum:]]\-\]*\]\).*\/1 2 == 3 == 4 5/' | sort -k 4 -g | sed 's/
== \(.*) ==\/1/' | sed 1,2d
2026/04/24 22:56:30 wazuh-logcollector: INFO: Monitoring full output of co
mmand(360): last -n 20
2026/04/24 22:56:30 wazuh-logcollector: INFO: (1950): Analyzing file: '/va
r/ossec/logs/active-responses.log'.
2026/04/24 22:56:30 wazuh-logcollector: INFO: (1950): Analyzing file: '/va
r/log/auth.log'.
2026/04/24 22:56:31 wazuh-logcollector: INFO: (1950): Analyzing file: '/va
r/log/syslog'.
2026/04/24 22:56:31 wazuh-logcollector: INFO: (1950): Analyzing file: '/va
r/log/dpkg.log'.
2026/04/24 22:56:31 wazuh-logcollector: INFO: (1950): Analyzing file: '/va
r/log/kern.log'.
2026/04/24 22:56:31 wazuh-modulesd:syscollector: INFO: Evaluation finished
.
2026/04/24 22:56:31 wazuh-logcollector: INFO: (1950): Analyzing file: '/va
r/log/suricata/eve.json'.
2026/04/24 22:56:31 wazuh-logcollector: INFO: Started (pid: 7250).
2026/04/24 22:56:39 wazuh-syscheckd: INFO: (6009): File integrity monitori
ng scan ended.
2026/04/24 22:56:39 wazuh-syscheckd: INFO: FIM sync module started.
2026/04/24 22:56:40 sca: INFO: Evaluation finished for policy '/var/ossec/
ruleset/sca/cis_ubuntu22-04.yml'
2026/04/24 22:56:40 sca: INFO: Security Configuration Assessment scan fini
shed. Duration: 11 seconds.
```

STEP 6: Generate a test alert (IMPORTANT)



To check suricata is running

```
test@OpenPLC-VM:~$ sudo suricata -D -c /etc/suricata/suricata.yaml -i ens33
25/4/2026 -- 05:48:00 - <Notice> - This is Suricata version 6.0.4 RELEASE running in SYSTEM mode
test@OpenPLC-VM:~$ ps aux | grep suricata
root      11775 12.2  2.1 630692 65352 ?        Ssl  02:34   23:49 /usr/bin/suricata -D --af-pac
ket -c /etc/suricata/suricata.yaml --pidfile /run/suricata.pid
root      14325 13.1  2.2 630692 67504 ?        Ssl  05:47    0:02 suricata -D -c /etc/suricata/
suricata.yaml -i ens33
test      14337  0.0  0.0   9212  2640 pts/0    S+   05:48    0:00 grep --color=auto suricata
test@OpenPLC-VM:~$
```